

Tangents and normals



1 $m = -\frac{1}{2}$

2 a $y = 5\pi x - 5\pi^2$

b $y = -3\sqrt{3}x + \frac{\sqrt{3}\pi}{6} + \frac{1}{2}$

3 a $x = \frac{\pi}{12}$ b $f'\left(\frac{\pi}{12}\right) = -4$ c $\theta = 104^\circ$

4 $\theta = 83^\circ$

5 a i $\frac{dy}{dx} = 2 \cos 2x$ ii Undefined
iii $x = \frac{\pi}{4}$

b i $\frac{dy}{dx} = \sqrt{2} \cos x$ ii -1
iii $y = -x + \frac{\pi}{4} + 1$

c i $\frac{dy}{dx} = -\sqrt{2} \sin x$ ii 1
iii $y = x - \frac{\pi}{4} + 1$

6 a $\frac{dy}{dx} = \frac{x \cos x - \sin x}{x^2}$ b $\frac{\pi^2}{4}$
c $y = \frac{\pi^2 x}{4} - \frac{\pi^3}{8} + \frac{2}{\pi}$

Trigonometric graphs

7 a $\frac{dy}{dx} = \frac{1}{1 + \cos x}$ b 0

8 $x = \underline{\underline{\frac{\pi}{4}}}, \underline{\underline{\frac{5\pi}{4}}}, \underline{\underline{\frac{9\pi}{4}}}, \underline{\underline{\frac{13\pi}{4}}}$



9

a $\left(\frac{\pi}{3}, -4\right), \left(\frac{4\pi}{3}, 4\right)$

b $f''(x) = 4 \sin\left(x + \frac{\pi}{6}\right)$

c $\left(\frac{\pi}{3}, -4\right)$ is a minimum, and $\left(\frac{4\pi}{3}, 4\right)$ is a maximum

10

a

i $y = 0, x = 0, \pi, 2\pi$

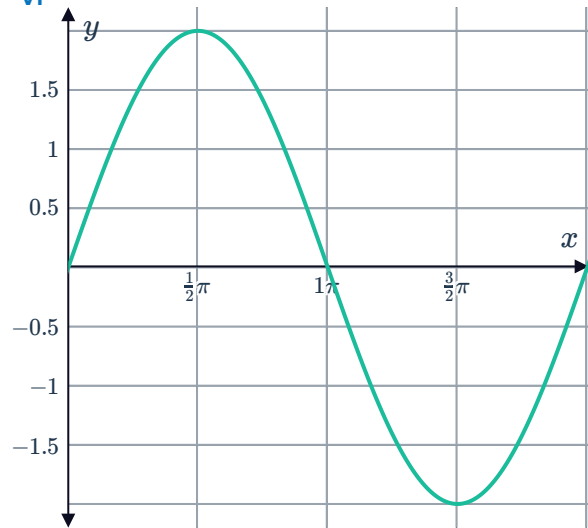
iii $\left(\frac{\pi}{2}, 2\right), \left(\frac{3\pi}{2}, -2\right)$

v Min at $\left(\frac{3\pi}{2}, -2\right)$, Max at $\left(\frac{\pi}{2}, 2\right)$

ii $f'(x) = 2 \cos x$

iv $f''(x) = -2 \sin x$

vi



b

i $y = 1, x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$



ii $f'(x) = -3 \sin 3x$

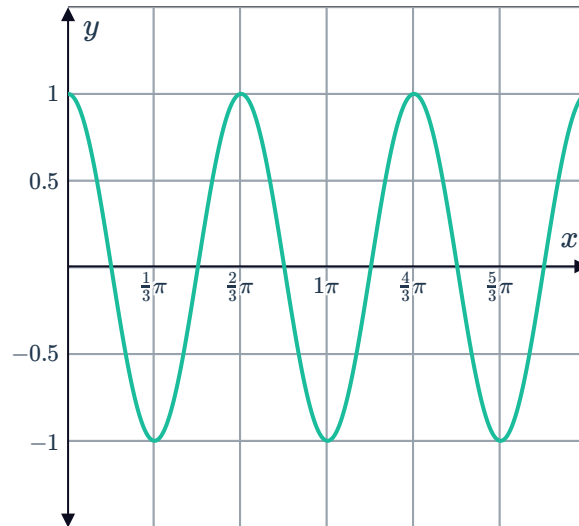
iii $\left(\frac{\pi}{3}, -1\right), (\pi, -1), \left(\frac{5\pi}{3}, -1\right), (0, 1), \left(\frac{2\pi}{3}, 1\right), \left(\frac{4\pi}{3}, 1\right), (2\pi, 1)$

iv $f''(x) = -9 \cos 3x$

v Minimum at $\left(\frac{\pi}{3}, -1\right), (\pi, -1), \left(\frac{5\pi}{3}, -1\right)$

Maximum at $(0, 1), \left(\frac{2\pi}{3}, 1\right), \left(\frac{4\pi}{3}, 1\right), (2\pi, 1)$

vi



c

i $y = 0, x = 0, \frac{\pi}{2}, \frac{3\pi}{2}$

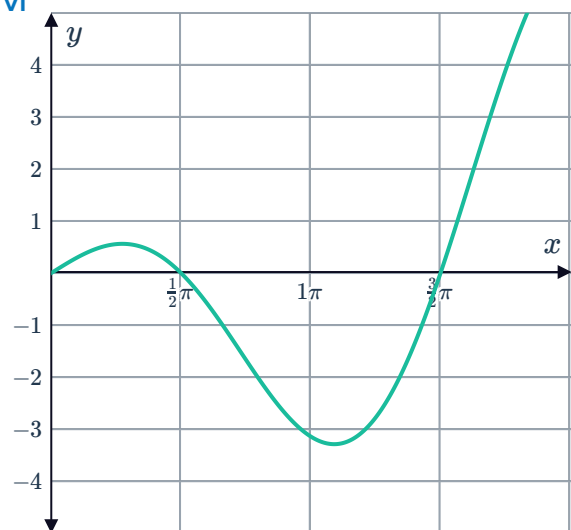
iii $(0.86, 0.56), (3.42, -3.29)$

v Minimum at $(3.42, -3.29)$
Maximum at $(0.86, 0.56)$

ii $f'(x) = \cos x - x \sin x$

iv $f''(x) = -2 \sin x - x \cos x$

vi



d

i $y = 1, x = \frac{\pi}{2}, \frac{3\pi}{2}$

iii $\left(\frac{3\pi}{4}, -\frac{e^{-\frac{3\pi}{4}}}{\sqrt{2}}\right)$

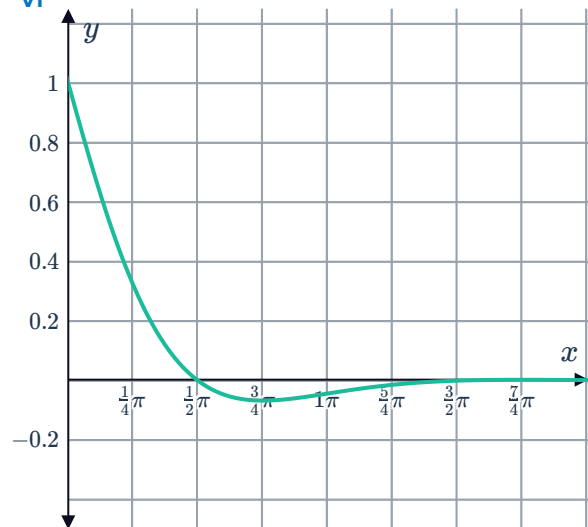
v Minimum point



ii $f'(x) = -e^{-x} \sin x - e^{-x} \cos x$

iv $f''(x) = 2e^{-x} \sin x$

vi



e

i $y = 0, x = 0, \pi, 2\pi$

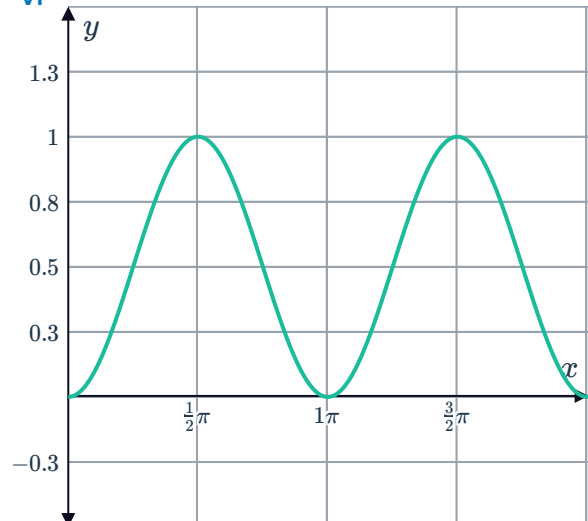
iii $(0, 0), (\pi, 0), (2\pi, 0), \left(\frac{\pi}{2}, 1\right), \left(\frac{3\pi}{2}, 1\right)$

v Minimum points: $(0, 0), (\pi, 0), (2\pi, 0)$
Maximum points: $\left(\frac{\pi}{2}, 1\right), \left(\frac{3\pi}{2}, 1\right)$

ii $f'(x) = 2 \sin x \cos x$

iv $f''(x) = 2(-\sin^2(x) + \cos^2(x))$

vi



f



i $y = 1, x = \frac{\pi}{2}, \frac{3\pi}{2}$

ii $f'(x) = -3 \cos^2(x) \sin x$

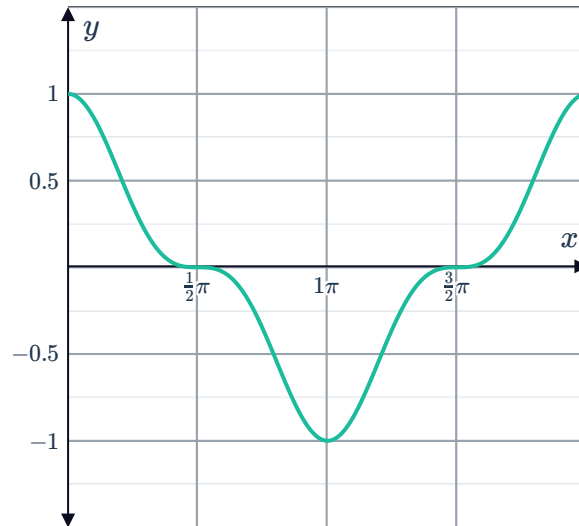
iii $(\pi, -1), (0, 1), (2\pi, 1), \left(\frac{\pi}{2}, 0\right), \left(\frac{3\pi}{2}, 0\right)$

iv $f''(x) = -3 \cos x (\cos^2(x) - 2 \sin^2(x))$

v Minimum point: $(\pi, -1)$, Maximum points: $(0, 1), (2\pi, 1)$

Horizontal points of inflection: $\left(\frac{\pi}{2}, 0\right), \left(\frac{3\pi}{2}, 0\right)$

vi



g

i $y = 0, x = 0$

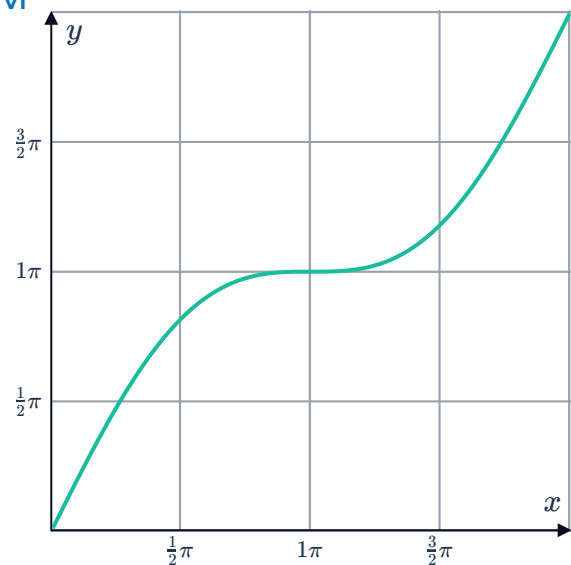
iii (π, π)

v Horizontal point of inflection

ii $f'(x) = 1 + \cos x$

iv $f''(x) = -\sin x$

vi





Applications

11 $k = \pm 5$

12

a It is 5 units to the left of the origin.

b $v(t) = 10 \sin 2t$

c No solution provided.

d -20 cm/s^2

13

a 0 cm

b $v(t) = -2 \cos^2(t) + \cos t - 4$

c -5 cm/s

14

a 0 cm

b 16 cm

c $v(t) = 48\pi \cos 3\pi t$

d $t = \frac{1}{6}, \frac{1}{2}$

e 16 cm

f -16 cm

g 32 cm

15

a $v(t) = 76\pi \cos 4\pi t$

b $76\pi \text{ cm/s}$

c $t = \frac{1}{4}, \frac{1}{2}$

d The centre of its swing at $x = 0$.